

Problem 1

Uniform distribution. A random variable that is equally likely to take any value in a finite set S is said to have the uniform distribution on S . If U is such a random variable and $\emptyset \neq R \subseteq S$, show that the distribution of U conditional on $\{U \in R\}$ is uniform on R .

Proof. Since U is uniformly distributed on set S , for any $s \in S$, we have:

$$P(U = s) = \frac{1}{|S|}$$

For any element $r \in R$, we need to compute the conditional probability $P(U = r \mid U \in R)$.

By the definition of conditional probability:

$$P(U = r \mid U \in R) = \frac{P(U = r \text{ and } U \in R)}{P(U \in R)}$$

Since $r \in R$, the event $\{U = r \text{ and } U \in R\}$ is equivalent to $\{U = r\}$. Therefore:

$$P(U = r \mid U \in R) = \frac{P(U = r)}{P(U \in R)}$$

We know that $P(U = r) = \frac{1}{|S|}$.

For $P(U \in R)$, we use the fact that U has a uniform distribution on S :

$$P(U \in R) = \sum_{t \in R} P(U = t) = \sum_{t \in R} \frac{1}{|S|} = \frac{|R|}{|S|}$$

Substituting these values:

$$P(U = r \mid U \in R) = \frac{\frac{1}{|S|}}{\frac{|R|}{|S|}} = \frac{1}{|R|}$$

Since this probability equals $\frac{1}{|R|}$ for every $r \in R$, the conditional distribution assigns equal probability to all elements in R . This is precisely the definition of a uniform distribution on R .

Therefore, the distribution of U conditional on $\{U \in R\}$ is uniform on R . $\square$

Problem 2

3. Let X be a random variable with distribution function

$$\mathbb{P}\{X \leq x\} = \begin{cases} 0 & \text{if } x \leq 0, \\ x & \text{if } 0 < x \leq 1, \\ 1 & \text{if } x > 1. \end{cases}$$

Let F be a distribution function which is continuous and strictly increasing. Show that $Y = F^{-1}(X)$ is a random variable having distribution function F .

Is it necessary that F be continuous and/or strictly increasing?

Proof. We need to show that if X has the given distribution function and F is continuous and strictly increasing, then $Y = F^{-1}(X)$ has distribution function F .

For any real number y , we have:

$$\begin{aligned} \mathbb{P}\{Y \leq y\} &= \mathbb{P}\{F^{-1}(X) \leq y\} \\ &= \mathbb{P}\{X \leq F(y)\} \end{aligned}$$

This equivalence holds because F is strictly increasing, so the inequality is preserved when applying F to both sides.

Since X has the distribution function:

$$\mathbb{P}\{X \leq x\} = \begin{cases} 0 & \text{if } x \leq 0 \\ x & \text{if } 0 < x \leq 1 \\ 1 & \text{if } x > 1 \end{cases}$$

Therefore:

$$\mathbb{P}\{X \leq F(y)\} = \begin{cases} 0 & \text{if } F(y) \leq 0 \\ F(y) & \text{if } 0 < F(y) \leq 1 \\ 1 & \text{if } F(y) > 1 \end{cases}$$

Since F is a distribution function, we know $0 \leq F(y) \leq 1$ for all y , which means we're always in the middle case where $\mathbb{P}\{X \leq F(y)\} = F(y)$. Thus, $\mathbb{P}\{Y \leq y\} = F(y)$, which proves that Y has distribution function F .

Regarding the necessity of the conditions:

Strict monotonicity: If F is not strictly increasing but merely non-decreasing, then for some values $a < b$ with $F(a) = F(b)$, the inverse function F^{-1} is not uniquely defined at $F(a)$. This makes $Y = F^{-1}(X)$ ambiguous.

Continuity: If F has jump discontinuities, then there exist values c in the range of F for which $F^{-1}(c)$ is not defined. If X takes such values with positive probability, then $Y = F^{-1}(X)$ is not properly defined as a random variable.

Both issues can be addressed by using the generalized inverse function:

$$F^{-1}(u) = \inf\{y : F(y) \geq u\}$$

With this definition, $Y = F^{-1}(X)$ will have distribution function F even when F is not continuous or strictly increasing, provided that F is at least right-continuous and non-decreasing (as any proper distribution function must be). $\square$

Problem 3

- (a) Show that any discrete random variable may be written as a linear combination of indicator variables.
- (b) Show that any random variable may be expressed as the limit of an increasing sequence of discrete random variables.
- (c) Show that the limit of any increasing convergent sequence of random variables is a random variable.

Proof. (a)

Let X be a discrete random variable taking values in the countable set $\{x_1, x_2, x_3, \dots\}$. We can express X as:

$$X = \sum_{i=1}^{\infty} x_i \mathbf{1}_{\{X=x_i\}} \quad (1)$$

where $\mathbf{1}_{\{X=x_i\}}$ is the indicator variable for the event $\{X = x_i\}$, defined as:

$$\mathbf{1}_{\{X=x_i\}}(\omega) = \begin{cases} 1 & \text{if } X(\omega) = x_i \\ 0 & \text{otherwise} \end{cases}$$

To verify this representation, consider any outcome ω in the sample space. If $X(\omega) = x_j$ for some j , then $\mathbf{1}_{\{X=x_i\}}(\omega) = 0$ for all $i \neq j$ and $\mathbf{1}_{\{X=x_j\}}(\omega) = 1$. Therefore:

$$\sum_{i=1}^{\infty} x_i \mathbf{1}_{\{X=x_i\}}(\omega) = x_j \cdot 1 + \sum_{i \neq j} x_i \cdot 0 = x_j = X(\omega)$$

Thus, X is indeed a linear combination of indicator variables.

(b)

Let X be any random variable. For each positive integer m , we partition the real line into intervals of the form $[k2^{-m}, (k+1)2^{-m})$ for $k \in \mathbb{Z}$ (i.e., all integers from $-\infty$ to ∞).

We define the discrete random variable X_m as:

$$X_m = \sum_{k=-\infty}^{\infty} k2^{-m} I_{k,m} \quad (2)$$

where $I_{k,m}$ is the indicator function of the event $\{k2^{-m} \leq X < (k+1)2^{-m}\}$.

This construction has the following properties:

1. X_m is a discrete random variable because it takes values in the countable set $\{k2^{-m} : k \in \mathbb{Z}\}$.
2. For any outcome ω , if $X(\omega) \in [k2^{-m}, (k+1)2^{-m})$ for some integer k , then $X_m(\omega) = k2^{-m}$, which is the left endpoint of the interval containing $X(\omega)$.
3. For each fixed ω , $X_m(\omega) \leq X(\omega)$ since $X_m(\omega) = k2^{-m}$ when $X(\omega) \in [k2^{-m}, (k+1)2^{-m})$.
4. As m increases, the partition becomes finer and $X_m(\omega)$ becomes a better approximation of $X(\omega)$. Specifically, if $X(\omega) \in [k2^{-m}, (k+1)2^{-m})$, then:

$$0 \leq X(\omega) - X_m(\omega) < (k+1)2^{-m} - k2^{-m} = 2^{-m} \quad (3)$$

5. The sequence $\{X_m(\omega)\}_{m=1}^{\infty}$ is increasing for each fixed ω . This is because as m increases, the left endpoints of the intervals in the partition either stay the same or increase to a value closer to $X(\omega)$.

Since $2^{-m} \rightarrow 0$ as $m \rightarrow \infty$, we have $X_m(\omega) \uparrow X(\omega)$ as $m \rightarrow \infty$ for all ω .

Therefore, any random variable X can be expressed as the limit of an increasing sequence of discrete random variables $\{X_m\}_{m=1}^{\infty}$.

(c)

Let $\{Y_n\}$ be an increasing sequence of random variables converging to Y . We need to show that Y is a random variable, which means we need to prove that for any Borel set B , the preimage $Y^{-1}(B) = \{\omega : Y(\omega) \in B\}$ is measurable.

Consider any $a \in \mathbb{R}$. We need to show that the set $\{\omega : Y(\omega) \leq a\}$ is measurable.

Since $\{Y_n\}$ is increasing and converges to Y , we have:

$$\{\omega : Y(\omega) \leq a\} = \{\omega : \lim_{n \rightarrow \infty} Y_n(\omega) \leq a\}$$

For an increasing sequence, $\lim_{n \rightarrow \infty} Y_n(\omega) \leq a$ if and only if $Y_n(\omega) \leq a$ for all n . Therefore:

$$\{\omega : Y(\omega) \leq a\} = \bigcap_{n=1}^{\infty} \{\omega : Y_n(\omega) \leq a\}$$

Since each Y_n is a random variable, each set $\{\omega : Y_n(\omega) \leq a\}$ is measurable. The countable intersection of measurable sets is measurable, so $\{\omega : Y(\omega) \leq a\}$ is measurable.

Since the preimage of any interval $(-\infty, a]$ under Y is measurable, and the Borel σ -algebra is generated by such intervals, Y is a random variable. $\square$

Problem 4

Express the distribution functions of

$$X^+ = \max\{0, X\}, \quad X^- = -\min\{0, X\}, \quad |X| = X^+ + X^-, \quad -X,$$

in terms of the distribution function F of the random variable X .

Proof. Let X be a random variable with distribution function $F(x) = \mathbb{P}(X \leq x)$. We'll express the distribution functions of the derived random variables in terms of F .

Distribution function of $X^+ = \max\{0, X\}$:

For $x < 0$, since $\max\{0, X\} \geq 0 > x$ for all values of X :

$$\mathbb{P}(X^+ \leq x) = 0$$

For $x \geq 0$:

$$\begin{aligned} \mathbb{P}(X^+ \leq x) &= \mathbb{P}(\max\{0, X\} \leq x) \\ &= \mathbb{P}(X \leq x) \\ &= F(x) \end{aligned}$$

Therefore:

$$\mathbb{P}(X^+ \leq x) = \begin{cases} 0 & \text{if } x < 0, \\ F(x) & \text{if } x \geq 0. \end{cases}$$

Distribution function of $X^- = -\min\{0, X\}$:

For $x < 0$, since $X^- \geq 0$ by definition:

$$\mathbb{P}(X^- \leq x) = 0$$

For $x \geq 0$, noting that $X^- = 0$ when $X \geq 0$ and $X^- = -X$ when $X < 0$:

$$\begin{aligned} \mathbb{P}(X^- \leq x) &= \mathbb{P}((X \geq 0) \cup (X < 0 \text{ and } -X \leq x)) \\ &= \mathbb{P}(X \geq 0) + \mathbb{P}(-x \leq X < 0) \end{aligned}$$

$$\begin{aligned}
&= 1 - \lim_{y \uparrow 0} F(y) + F(0) - F(-x) \\
&= 1 - \lim_{y \uparrow -x} F(y)
\end{aligned}$$

Therefore:

$$\mathbb{P}(X^- \leq x) = \begin{cases} 0 & \text{if } x < 0, \\ 1 - \lim_{y \uparrow -x} F(y) & \text{if } x \geq 0. \end{cases}$$

Distribution function of $|X| = X^+ + X^-$:

For $x < 0$, since $|X| \geq 0$:

$$\mathbb{P}(|X| \leq x) = 0$$

For $x \geq 0$, using $|X| = X$ when $X \geq 0$ and $|X| = -X$ when $X < 0$:

$$\begin{aligned}
\mathbb{P}(|X| \leq x) &= \mathbb{P}(0 \leq X \leq x) + \mathbb{P}(-x \leq X < 0) \\
&= F(x) - F(0) + F(0) - F(-x) \\
&= F(x) - \lim_{y \uparrow -x} F(y)
\end{aligned}$$

Therefore:

$$\mathbb{P}(|X| \leq x) = \begin{cases} 0 & \text{if } x < 0, \\ F(x) - \lim_{y \uparrow -x} F(y) & \text{if } x \geq 0. \end{cases}$$

Distribution function of $-X$:

For any real x :

$$\begin{aligned}
\mathbb{P}(-X \leq x) &= \mathbb{P}(X \geq -x) \\
&= 1 - \mathbb{P}(X < -x) \\
&= 1 - \lim_{y \uparrow -x} F(y)
\end{aligned}$$

□

Problem 5

The real number m is called a median of the distribution function F whenever $\lim_{y \rightarrow m} F(y) \leq \frac{1}{2} \leq F(m)$.

(a) Show that every distribution function F has at least one median, and that the set of medians of F is a closed interval of $\mathbb{R}$.

(b) Show, if F is continuous, that $F(m) = \frac{1}{2}$ for any median m .

Proof. (a)

Define $m = \sup\{x : F(x) < \frac{1}{2}\}$. We will show that m is a median.

For any $y < m$, by the definition of supremum, we have $F(y) < \frac{1}{2}$. This implies that $\lim_{y \uparrow m} F(y) \leq \frac{1}{2}$.

Now we need to show that $F(m) \geq \frac{1}{2}$. Suppose, for contradiction, that $F(m) < \frac{1}{2}$. Since F is right-continuous, there exists some $\delta > 0$ such that $F(m + \epsilon) < \frac{1}{2}$ for all $0 < \epsilon < \delta$. But this means that $m + \frac{\delta}{2}$ belongs to the set $\{x : F(x) < \frac{1}{2}\}$, which contradicts m being the supremum of this set. Therefore, $F(m) \geq \frac{1}{2}$.

Thus, m satisfies $\lim_{y \uparrow m} F(y) \leq \frac{1}{2} \leq F(m)$, making it a median.

Similarly, define $M = \sup\{x : F(x) \leq \frac{1}{2}\}$. Using a similar argument, we can show that M is also a median.

For any $x \in [m, M]$, we have:

$$\lim_{y \uparrow x} F(y) \leq \lim_{y \uparrow m} F(y) \leq \frac{1}{2}$$

because F is non-decreasing. Also:

$$F(x) \geq \frac{1}{2}$$

because $x \geq m$ and F is non-decreasing. Therefore, every $x \in [m, M]$ is a median.

Conversely, if $x < m$, then $F(x) < \frac{1}{2}$, so x cannot be a median. If $x > M$, then by definition of M , we have $F(x) > \frac{1}{2}$, and by right-continuity of F , we get $\lim_{y \uparrow x} F(y) > \frac{1}{2}$, so x cannot be a median.

Therefore, the set of all medians is exactly the closed interval $[m, M]$.

(b)

Let m be any median of the continuous distribution function F . By definition of median:

$$\lim_{y \uparrow m} F(y) \leq \frac{1}{2} \leq F(m)$$

Since F is continuous at m , we have $\lim_{y \uparrow m} F(y) = F(m)$. The inequality above becomes:

$$F(m) \leq \frac{1}{2} \leq F(m)$$

This can only be satisfied when $F(m) = \frac{1}{2}$.

Therefore, when F is continuous, every median m satisfies $F(m) = \frac{1}{2}$. □

Problem 6

Let $(\Omega, \mathcal{F})$ be a measurable space, $A_1, \dots, A_n \in \mathcal{F}$ and $b_1, \dots, b_n \in \mathbb{R}$. Set $X = b_1 1_{A_1} + \dots + b_n 1_{A_n}$. Suppose that for all $i \neq j$, $A_i \cap A_j = \emptyset$ and $b_i \neq b_j$. Show that $\sigma(X) = \sigma(\{A_1, \dots, A_n\})$.

Proof. Let $(\Omega, \mathcal{F})$ be a measurable space, $A_1, \dots, A_n \in \mathcal{F}$ be disjoint sets and $b_1, \dots, b_n \in \mathbb{R}$ be distinct real numbers. We define $X = \sum_{i=1}^n b_i 1_{A_i}$.

We want to show that $\sigma(X) = \sigma(\{A_1, \dots, A_n\})$.

$\Rightarrow \sigma(X) \subseteq \sigma(\{A_1, \dots, A_n\})$:

For any Borel set $B \in \mathcal{B}(\mathbb{R})$, we need to show that $X^{-1}(B) \in \sigma(\{A_1, \dots, A_n\})$.

Note that X takes only the values from the set $\{b_1, \dots, b_n, 0\}$, where 0 is taken when $\omega \notin \bigcup_{i=1}^n A_i$. Let's define $A_0 = \Omega \setminus \bigcup_{i=1}^n A_i$ and $b_0 = 0$ for convenience.

For any $\omega \in \Omega$, there is a unique index $j \in \{0, 1, \dots, n\}$ such that $\omega \in A_j$, and thus $X(\omega) = b_j$.

Therefore,

$$\begin{aligned} X^{-1}(B) &= \{\omega \in \Omega : X(\omega) \in B\} \\ &= \{\omega \in \Omega : \exists j \in \{0, 1, \dots, n\} \text{ such that } \omega \in A_j \text{ and } b_j \in B\} \\ &= \bigcup_{j=0}^n \{A_j : b_j \in B\} \end{aligned}$$

Since each $A_j \in \sigma(\{A_1, \dots, A_n\})$ (note that A_0 can be expressed as $\Omega \setminus \bigcup_{i=1}^n A_i$, which is in $\sigma(\{A_1, \dots, A_n\})$), we have $X^{-1}(B) \in \sigma(\{A_1, \dots, A_n\})$.

Thus, $\sigma(X) \subseteq \sigma(\{A_1, \dots, A_n\})$.

$\Leftarrow \sigma(\{A_1, \dots, A_n\}) \subseteq \sigma(X)$:

We need to show that for each $i \in \{1, \dots, n\}$, $A_i \in \sigma(X)$.

Since the values $b_1, \dots, b_n$ are distinct, for each $i \in \{1, \dots, n\}$, the set $\{b_i\}$ is a Borel set in $\mathbb{R}$. We have:

$$\begin{aligned} X^{-1}(\{b_i\}) &= \{\omega \in \Omega : X(\omega) = b_i\} \\ &= \{\omega \in \Omega : \sum_{j=1}^n b_j 1_{A_j}(\omega) = b_i\} \end{aligned}$$

For any $\omega \in A_i$, we have $X(\omega) = b_i$ because $A_i \cap A_j = \emptyset$ for $i \neq j$. For any $\omega \notin A_i$, we have $X(\omega) \neq b_i$ because the values $b_1, \dots, b_n$ are distinct and the sets $A_1, \dots, A_n$ are disjoint.

Therefore, $X^{-1}(\{b_i\}) = A_i$, which implies $A_i \in \sigma(X)$ for each $i \in \{1, \dots, n\}$.

Hence, $\sigma(\{A_1, \dots, A_n\}) \subseteq \sigma(X)$.

So we conclude that $\sigma(X) = \sigma(\{A_1, \dots, A_n\})$. □

Problem 7

Consider the random walk on $\{0, 1, \dots, N\}$ that reflects at 0, introduced in the class. More precisely, let $S_0 = k, S_{j+1} = S_j + X_j$, where $X_j = 1$ if $S_j = 0$, otherwise $\mathbb{P}[X_j = 1] = p$ and $\mathbb{P}[X_j = -1] = q$. Compute the expected number of steps to first reach state N , as a function of k . Distinguish the cases $p = q$ and $p \neq q$.

Proof. Random walk on $\{0, 1, \dots, N\}$ with:

$$\begin{aligned}\mathbb{P}[X_j = 1 \mid S_j = 0] &= 1 \text{ (reflection at 0)} \\ \mathbb{P}[X_j = 1 \mid S_j \neq 0] &= p \\ \mathbb{P}[X_j = -1 \mid S_j \neq 0] &= q = 1 - p\end{aligned}$$

Let h_k be the expected number of steps to first reach state N from initial state k .

Recursive equations:

$$\begin{aligned}h_k &= 1 + ph_{k+1} + qh_{k-1}, \quad 0 < k < N \\ h_0 &= 1 + h_1 \\ h_N &= 0\end{aligned}$$

Case 1: Symmetric random walk ($p = q = 1/2$)

For $0 < k < N$, the recursive equation becomes:

$$\begin{aligned}h_k &= 1 + \frac{1}{2}h_{k+1} + \frac{1}{2}h_{k-1} \\ \Rightarrow h_{k+1} - 2h_k + h_{k-1} &= -2\end{aligned}$$

This second-order linear difference equation has general solution:

$$h_k = A + Bk + k(N - k)$$

From boundary conditions:

$$\begin{aligned}h_N = 0 &\Rightarrow A + BN = 0 \\ h_0 = 1 + h_1 &\Rightarrow B = -1 \Rightarrow A = N\end{aligned}$$

Therefore:

$$h_k = (N - k)(k + 1)$$

Case 2: Asymmetric random walk ($p \neq q$)

Rearranging the recursive equation:

$$ph_{k+1} - h_k + qh_{k-1} = -1$$

Let $\rho = \frac{q}{p}$. The general solution has form:

$$h_k = A + B\rho^k + \frac{k}{q-p} + \frac{q}{q-p}$$

Applying boundary conditions:

$$h_N = 0$$

$$h_0 = 1 + h_1$$

Solving for constants A and B :

$$A = \frac{1}{p-q} \left(\frac{1-\rho^N}{1-\rho} - N \right)$$

$$B = \frac{1}{p-q} \cdot \frac{1}{1-\rho^N}$$

Final solution when $p \neq q$:

$$h_k = \begin{cases} \frac{N-k}{p-q} & \text{if } p > q \\ \frac{1}{p-q} \left[\frac{1-\rho^N}{1-\rho} - N - \frac{\rho^k - \rho^N}{1-\rho^N} \right] & \text{if } p < q \end{cases}$$

□

Problem 8

If $F : \mathbb{R} \rightarrow [0, 1]$ satisfies monotone increasing, right continuity, $\lim_{x \rightarrow -\infty} F(x) = 0$ and $\lim_{x \rightarrow +\infty} F(x) = 1$, show that F is the distribution function of some random variable on some probability space. (Hint: you may want to revisit Section 2.3, exercise 3)

Proof. Let $F : \mathbb{R} \rightarrow [0, 1]$ satisfy:

- Monotone increasing
- Right continuous
- $\lim_{x \rightarrow -\infty} F(x) = 0$
- $\lim_{x \rightarrow +\infty} F(x) = 1$

We construct a random variable with distribution function F using the hint that the limit of any increasing sequence of random variables is a random variable.

Let $(\Omega, \mathcal{F}, \mathbb{P}) = ((0, 1], \mathcal{B}((0, 1]), \lambda)$ where λ is Lebesgue measure.

For each $n \in \mathbb{N}$, define:

$$X_n(\omega) = \inf\{x = k2^{-n} : F(x) \geq \omega, k \in \mathbb{Z}\}$$

The sequence $\{X_n\}$ is decreasing: $X_n(\omega) \geq X_{n+1}(\omega)$ for all $\omega \in \Omega$.

Define $X(\omega) = \lim_{n \rightarrow \infty} X_n(\omega)$. By the hint, X is a random variable since it's the limit of a monotone sequence of random variables.

From the construction and right continuity of F :

$$X(\omega) = \inf\{x \in \mathbb{R} : F(x) \geq \omega\}$$

For any $a \in \mathbb{R}$:

$$\begin{aligned} \mathbb{P}(X \leq a) &= \mathbb{P}(\{\omega \in \Omega : F(a) \geq \omega\}) \\ &= F(a) \end{aligned}$$

Therefore, F is the distribution function of the random variable X . □